CSE-2006
MicroProcessors and Interfacing

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S.no	Component	Specification	Justification
1	Arduino Uno	• Microcontroller: ATmega328P • Operating Voltage: 5V • Input Voltage (recommended): 7-12V • Inout Voltage (limit): 6-20V • Digital I/O Pins: 14 (of which 6 provide PWM output) • PWM Digital I/O Pins: 6 • Analog Input Pins: 6 • LED_BUILTIN: 13	It is the main microcontroller that will be connected to the various components at hand via its various pins. It can be connected to the computer with a simple USB cable.
2	LED dot matrix	• LED colour: RGB. • Comm Pin -cathode pins. • Number of LED's per board: 64. • LED layout: 8 x 8. • LED diameter: 5mm. • Forward voltage: 2V. • Forward current: 5 – 10mA. • Board Dimensions: 60 x 60 mm.	Main output LED that showcases the emotion emoticon by using a circular face. If plant is happy, it shows a happy face or else, a sad face.

S.no	Component	Specification	Justification
3	Soil Moisture Sensor	• Analogue Data range : 0 - 1024 • Detection length : 38 mm • Working Voltage : 2V-5V	For a plant, we require a detection length that is fairly shallow to understand the moisture levels of the soil
4	DHT11 Temperature and Humidity sensor	• Humidity measuring range: 20% - 90% • Temperature measuring range: 0-50 degrees • Weight: 0.02 Kg	The temperature and humidity is extremely vital for the plant's survival and hence has to be measured with accuracy.
5	LCD display	• Dimensions: 16 x 2 • Pixels: 1280 (32x40) • Characters: 32 (16x2) • Pixels per character: 40 (5x8)	It displays the level of humidity, temperature, overall score and sunlight.
6	NodeMCU	Operating Voltage - 3.3V Analog inout pins - 1 Digital I/O pins - 16	It is used to connect to WIFI, used in our project as network connectivity module.
7	Piezo buzzer	Frequency range is 3300Hz Operating voltage - 3V to 24V DC Operating Temperature is -20 to 60 celsius	Used to buzz.

Software Requirements

1. Arduino IDE 2.0.0 

   The open-source Arduino Software (IDE) to write code and upload it to the board.

2. Thinkercad 

   Tinkercad is a free web app for 3D design, electronics, and coding. For making the digital model of the project

Arduino Code

```
#include <Wire.h>                                   //  Including Wire library for the I2C Connected to the LCD Display
#include <LiquidCrystal_I2C.h>                      //  Including Liquid crystal library for I2C
#include "DHT.h"                                    //  Including DHT library for DHT11 Temperature and Humidity Sensor
#include <LedControl.h>                             //  Including LED library for controlling 8X8 dot matrix using MAX7219
#define DHTPIN 7                                    //  Defining the sensor pin of DHT11 sensor connected to Arduino
#define DHTTYPE DHT11                               // Defining the type of DHT sensor, since this library is built also for DHT22


const int en = 2, rw = 1, rs = 0, d4 = 4, d5 = 5, d6 = 6, d7 = 7, bl = 3;      //LCD pinouts for the I2C
const int i2c_addr = 0x27;                          //I2C Address, found using another programme
const int msensor = A1;                             //Moisture Sensor data pin Connected to analog pin A1 in Arduino
const int Vin=A0;                                   //Pin to receive analog readings from the photoresistor
const int DIN = 10;                                 //Data pin of the MAX7219 8X8 dot matrix controller
const int CS =  9;                                  //Another Data pin of the MAX7219 8X8 dot matrix controller
const int CLK = 8;                                  //Clock Data pin of the MAX7219 8X8 dot matrix controller


LedControl lc=LedControl(DIN,CLK,CS,0);             //Defininng pins to control the 8X8 dot matrix
LiquidCrystal_I2C lcd(i2c_addr, en, rw, rs, d4, d5, d6, d7, bl, POSITIVE);  //Defining LCD connections to I2C bus
DHT dht(DHTPIN, DHTTYPE);                           //Defining and Setting up DHT sensor

int msvalue = 0;                                    //Initial Value of Moisture Sensor
int light= 0;                                       //Initial value of voltage across photoresistor
float tempC=0.0;                                    //Assigning variable to calculate the Temperature score
float humC=0.0;                                     //Assigning variable to calculate the Humidity score
float msenC=0.0;                                    //Assigning variable to calculate the Moisture score
float lightC=0.0;                                   //Assigning variable to calculate the Sunlight score
float hum;                                          //Stores humidity value in percent
float temp;                                         //Stores temperature value in Celcius
byte smile[8]=   {0x3C,0x42,0xA5,0x81,0xA5,0x99,0x42,0x3C};     //Byte array to store the "Smile Expression"
byte neutral[8]= {0x3C,0x42,0xA5,0x81,0xBD,0x81,0x42,0x3C};     //Byte array to store the "Neutral Expression"
byte sad[8]=   {0x3C,0x42,0xA5,0x81,0x99,0xA5,0x42,0x3C};       //Byte array to store the "Sad Expression"

void setup()
 {
   Serial.begin(9600);                              //Setting up serial monitor
   pinMode(msensor, INPUT);                         //giving analog pin address (A1) to Arduino for Moisture sensor
```

```
  pinMode(Vin,INPUT);                          //giving analog pin address (A2) to Arduino for photoresistor
  lcd.begin(16,2);                                                  //Set display type as 16 char, 2 rows
  dht.begin();                                                                        //Initialize DHT-22
  lc.shutdown(0,false);                                  //Initializing the MAX7219 8X8 dot matrix controller
  lc.setIntensity(0,2);                                    //Toggle to adjust LED brightness, maximum is 15
  lc.clearDisplay(0);                                    //Clearing any previous display from the dot matrix
  lcd.setCursor(0,0);
  delay(2000);
  lcd.print("    Welcome");                                                          //Intro lines
  lcd.setCursor(0,1);
  lcd.print("      to");                                                             //Intro lines
  delay(2000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("    Plant");                                                          //Intro lines
  lcd.setCursor(0,1);
  lcd.print("  Communicator");                                                     //Intro lines
  delay(2000);
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("      By");                                                           //Intro lines
  lcd.setCursor(0,1);
  lcd.print("  Sensors 2355");                                                     //Intro lines
  delay(2000);
  lcd.clear();

 }
void loop()
{
 hum = dht.readHumidity();                                                  //Get Humidity value
 temp= dht.readTemperature();                                            //Get Temperature value
 float score=0.0;                                          //Assigning a variable to get 'Care taking' score

 lcd.clear();                                                                        //Clear the display
                                                                                  //Temperature Parameters
 if(temp>=18 && temp<=26)
 {
  lcd.setCursor(0,0);
  lcd.print("Temp:  ");
  lcd.print(temp);
  lcd.print(" C");
  lcd.setCursor(0,1);
  lcd.print("   PERFECT");
  tempC=3.0;
  delay(2000);
 }
```

```
else if(temp>=6 || temp<=34)
{
  lcd.setCursor(0,0);
  lcd.print("Temp:  ");
  lcd.print(temp);
  lcd.print(" C");
  lcd.setCursor(0,1);
  lcd.print("      FINE");
  tempC=2.0;
  delay(2000);
}
else if((temp<6 || temp>34))
```

```
{
  lcd.setCursor(0,0);
  lcd.print("Temp:  ");
  lcd.print(temp);
  lcd.print(" C");
  lcd.setCursor(0,1);
  lcd.print("     BAD");
  tempC=1.0;
  delay(2000);
}
lcd.clear();
                                                    //Humidity Parameters

if(hum>=50 && hum<=60)
 {
  lcd.setCursor(0,0);
  lcd.print("Humidity: ");
  lcd.print(hum);
  lcd.print("%");
  lcd.setCursor(0,1);
  lcd.print("    PERFECT");
  humC=3.0;
  delay(2000);
}
else if(hum>=40 || hum<=80)
{
  lcd.setCursor(0,0);
  lcd.print("Humidity: ");
  lcd.print(hum);
  lcd.print("%");
  lcd.setCursor(0,1);
  lcd.print("     FINE");
  humC=2.0;
  delay(2000);
```

```
}
else if(hum>=40 || hum<=80)
{
  lcd.setCursor(0,0);
  lcd.print("Humidity: ");
  lcd.print(hum);
  lcd.print("%");
  lcd.setCursor(0,1);
  lcd.print("      FINE");
  humC=2.0;
  delay(2000);
}
else if((hum<40 || hum>80))
{
  lcd.setCursor(0,0);
  lcd.print("Humidity: ");
  lcd.print(hum);
  lcd.print("%");
  lcd.setCursor(0,1);
  lcd.print("      BAD");
  humC=1.0;
  delay(2000);
}
lcd.clear();
                                                                    //Soil Moisture Parameters

msvalue = analogRead(msensor);//Reading Value from Moisture Sensor          //Get Soil Moisture values from analog pin

if(msvalue>=200 && msvalue<=600)
{
  lcd.setCursor(0,0);
  lcd.print(" Soil Moisture");
  lcd.setCursor(0,1);
  lcd.print("    PERFECT");
  msenC=3.0;
```

```
  delay(2000);
}
else if(msvalue>600)
{
  lcd.setCursor(0,0);
  lcd.print(" Soil Moisture");
  lcd.setCursor(0,1);
  lcd.print("     LOW");
  msenC=1.0;
  delay(2000);
}
else if(msvalue<200)
{
  lcd.setCursor(0,0);
  lcd.print(" Soil Moisture");
  lcd.setCursor(0,1);
  lcd.print("Excess Water");
  msenC=2.0;
  delay(2000);
}
lcd.clear();
                                                    //Sunlight Parameters

light = analogRead(Vin);                  //Reading Voltage across the photoresistor to calculate the light intensity
Serial.println(Vin);
if(light>=750)
{
  lcd.setCursor(0,0);
  lcd.print("   Sunlight");
  lcd.setCursor(0,1);
  lcd.print("   PERFECT");
  lightC=3.0;
  delay(2000);
}
```

```
else if(light>=300)
{
  lcd.setCursor(0,0);
  lcd.print("    Sunlight");
  lcd.setCursor(0,1);
  lcd.print("Could be Better");
  lightC=2.0;
  delay(2000);
}
else if(light<300)
{
  lcd.setCursor(0,0);
  lcd.print("   Sunlight");
  lcd.setCursor(0,1);
  lcd.print("Needs Sunlight!!");
  lightC=1.0;
  delay(2000);
}
lcd.clear();

                                        //All parameters have been checked and we now calculate the score
score=((tempC+humC+msenC+lightC)/4);
if(score>=2.5)
{
  lcd.setCursor(0,0);
```

```
  lcd.print("Score: ");
  lcd.print(score);
  lcd.print("/3.00");
  lcd.setCursor(0,1);
  lcd.print(" Plant is HAPPY                    //Print how the plant feels on bottom line
  printByte(smile);                             //Display Plant's expression on the DOT Matrix
  delay(3000);
 }
 else if(score>=2)
 {
  lcd.setCursor(0,0);                           //Print Caretaking score on top line
  lcd.print("Score: ");
  lcd.print(score);
  lcd.print("/3.00");
  lcd.setCursor(0,1);                           //Print how the plant feels on bottom line
  lcd.print(" Plant is Fine");                  //Display Plant's expression on the DOT Matrix
  printByte(neutral);
  delay(3000);
 }
 else if(score>=1)
 {
  lcd.setCursor(0,0);                           //Print Caretaking score on top line
  lcd.print("Score: ");
  lcd.print(score);
  lcd.print("/3.00");
  lcd.setCursor(0,1);                           //Print how the plant feels on bottom line
  lcd.print(" Plant is SAD");                   //Display Plant's expression on the DOT Matrix
  printByte(sad);
  delay(3000);
 }
 lcd.clear();
}

void printByte(byte character [])               //Function, called to print Expressions on the DOT Matrix Display
{
 int i = 0;
 for(i=0;i<8;i++)
 {
   lc.setRow(0,i,character[i]);
 }
}
```

****

Assembled Hardware

